

Lab 4 - Digital Sampling

<https://youtu.be/vDQibxb5m4A>

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A02-X

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Introduction

The purpose of lab 4, Digital Sampling, is to convert mp3 waveforms into analog electrical signals, utilize a personal DAQ & microphone setup to record sound waves, and acquire digital data of time-varying signals. In this experiment, a LabJack T4 and the LabVIEW software served as the DAQ.

♦ Lab 4 can be broken down into 2 major sections:

Week 2

- Utilize DAQ to analyze and identify waveforms from 11 different audio tracks
- Interpret power spectra of complex waveforms
- Examine quantization error
- Explore the effects of record length and sampling rate on power spectrum
- Observe aliasing
- Examine noise
- Explore the implementation of a low pass filter

Week 3

- Conduct your own experiment



Week 2 – Results

Track	Signal Type	Peak Frequencies (Hz)	Relative Power Level (Vrms/ $\sqrt{\text{Hz}}$)
1	Single Sine Wave	97.66	0.149
2	Single Sine Wave	1000.98	0.147
3	Triangle Wave	1000.98, 3000, 5000, 9000, 11000	0.123, 0.014, 0.007, 0.005, 0.004, 0.003
4	Square Wave*	1000, 2k, 3k, 4k, 5k, 6k, 7k, 8, 9k, 10k, 11k ,12k	0.155, 0.008, 0.065, 0.013, 0.037, 0.010, 0.023, 0.015, 0.030, 0.022, 0.030, 0.026
5	Single Sine Wave	3002.93	0.199
6	Ramp Wave (Noisy)*	1000, 2k, 3k, 4k, 5k, 6k, 7k, 8, 9k, 10k, 11k ,12k	0.087, 0.048, 0.036, 0.019, 0.015, 0.020, 0.026, 0.027, 0.027, 0.027, 0.026, 0.025
7	Single Sine Wave (Noisy)	1000.98	0.141
8	Single Sine Wave (Very Noisy)	1000.98	0.06474
9	Sum of 3 Sine Wave	100, 1000, 10009.77	0.053, 0.046, 0.094
10	Product of 2 Sine Wave	900, 1100	0.101, 0.102
11	Sine Wave of a Sine Wave**	0 – 1050	0.173



Week 2 – Results

Interpret power spectra of complex waveforms

Time History Frequency: 200 Hz, 1000 Hz. Power Spectrum Frequency: 900 Hz, 1100 Hz

$$\sin(2\pi(900)t) + \sin(2\pi(1100)t) = 2 \sin(2\pi(1000)t) \cos(2\pi(200)t)$$

Compared to the triangle wave, the square wave has higher power spectrum peaks at higher frequencies.

Examine quantization error

When the amplifier is lowered, the square wave loses its shape and tends to act more like a single sine wave.

Explore the effects of record length and sampling rate on power spectrum

Record Length – Changes the number of distinct points on the power spectrum

Sampling Rate – Changes the frequency resolution of the power spectrum.

Observe Aliasing

At low sampling rates, power spectrum readings become significantly skewed to the left.

Examine Noise

White noise was present in three samples & Track 7 had a higher signal-to-noise ratio than Track 8.



Explore the implementation of a low pass filter

It is possible to filter out higher frequencies while keeping lower frequencies the same with the implementation of a cutoff frequency.



Week 3 – Purpose & Motivation

Background: In high school, I enjoyed playing the trombone. Oftentimes, I would need to record myself playing for either assignments or auditions. Over time, I began to notice that these online recordings generally sounded different than how I played the instrument in real life.

Main Questions: Why do the same musical notes sound different on video compared to real life? What exactly happens when a sound wave is recorded and played back as a digital signal?

Purpose: The purpose of my Week 3 experiment is to analyze the differences in power spectra between music notes recorded by the audio jack and the microphone. The audio jack serves to represent that digital signal stored in the audio output device, while the microphone serves to represent the real-life sound wave create by the audio output device.

Hypothesis: If an audio jack were to record data on a particular music note, then its corresponding power spectrum would be significantly more compressed compared to that recorded by a microphone. 



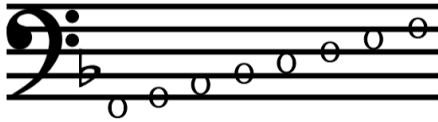
Equipment

- ◆ LabJack T4 DAQ
- ◆ USB-A to USB-B Power Cable
- ◆ LabJack Screwdriver
- ◆ Miniature Eyeglass Screwdriver
- ◆ Microphone
- ◆ 3.5mm Audio Breakout Adapter
- ◆ 3male-male Jumper Wires
- ◆ Audio Output Device (Phone/Computer/Etc.)
- ◆ Notebook

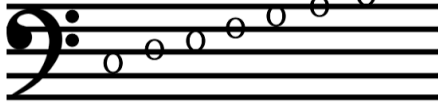


Procedures

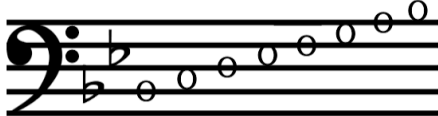
F Major Scale



C Major Scale



Bb Major Scale



1. Setup Tone Generator
 - Find device with wired audio output jack, set volume to 100%, and open the following tone generator:

<https://www.szynalski.com/tone-generator/>

2. Setup DAQ, Audio Jack, and Function Generator
 - Connect DAQ to computer using the USB cable
 - Connect audio output of the tone generator to the DAQ utilizing the 3.5mm audio jack and 2 jumper wires
 - Open GTAE Simple DAQ application

3. Validate the Audio Jack
 - Set tone generator to output the following tones: 200 Hz, 500 Hz, 1000Hz, 10000 Hz
 - Set Sampling Rate and Record Length to appropriate values
 - Record this tone by clicking Enable and waiting 3 seconds before clicking Disable
 - Repeat this procedure for each frequency and ensure consistent findings with the GTAE Simple DAQ application

4. Analyze and Record Audio Jack Data

Set the tone generator to the base note of the C major scale

Record this note by clicking Enable and waiting 3 seconds before clicking Disable

Using the data gathered, plot the power spectrum of the note in Excel and identify the peak frequency and power level

Repeat this procedure for every note in the C major scale

Repeat and adapt this procedure for both the F and Bb major scales to better ensure accuracy

5. Setup and Validate the Microphone

- Remove the 3.5mm audio jack before connect three jumper wires to the screw terminal on the microphone and to the GND, VS, and AIN0 on the DAQ
- Set tone generator to output the following tones: 200 Hz, 500 Hz, 1000Hz, 10000 Hz
- Set Sampling Rate and Record Length to appropriate values
- Record this tone by clicking Enable and waiting 3 seconds before clicking Disable
- Repeat this procedure for each frequency and ensure consistent findings with the GTAE Simple DAQ application

6. Analyze and Record Microphone Data

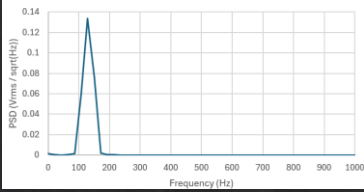
- Set the tone generator to the base note of the C major scale
- Record this note by clicking Enable and waiting 3 seconds before clicking Disable
- Using the data gathered, plot the power spectrum of the note in Excel and identify the peak frequency and power level
- Repeat this procedure for every note in the C major scale
- Repeat and adapt this procedure for both the F and Bb major scales to better ensure accuracy



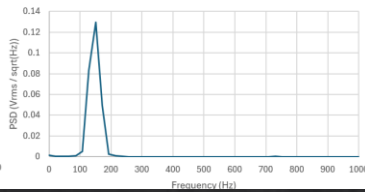
Frequency (Hz)
Power Level (Vrms/ $\sqrt{\text{Hz}}$)

Week 3 Results – C Major Scale

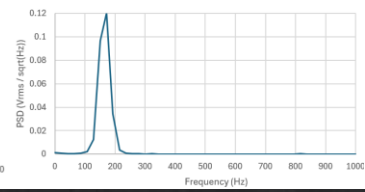
Audio Jack C3 Power Spectrum



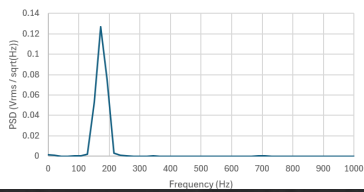
Audio Jack D3 Power Spectrum



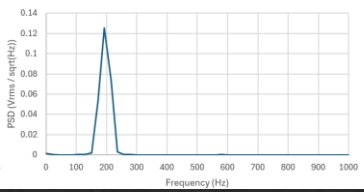
Audio Jack E3 Power Spectrum



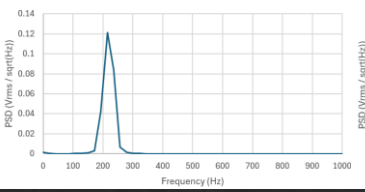
Audio Jack F3 Power Spectrum



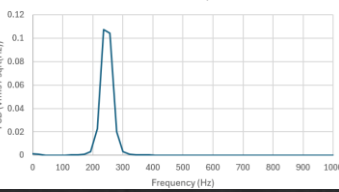
Audio Jack G3 Power Spectrum



Audio Jack A3 Power Spectrum



Audio Jack B3 Power Spectrum



Frequency: 128.91
Power Level: 0.134
C3

Frequency: 107.42,
257.81, 408.20
Power Level: 0.010,
0.030, 0.003

Frequency: 150.39
Power Level: 0.130
D3

Frequency: 150.29,
300.78, 451.17
Power Level: 0.011,
0.012, 0.003

Frequency: 171.88
Power Level: 0.120
E3

Frequency: 171.88,
322.27, 494.43
Power Level: 0.027,
0.009, 0.004

Frequency: 171.88
Power Level: 0.127
F3

Frequency: 171.88,
343.75, 515.63
Power Level: 0.035,
0.019, 0.003

Frequency: 193.36
Power Level: 0.125
G3

Frequency: 193.36,
386.72, 580.08
Power Level: 0.047,
0.005, 0.007

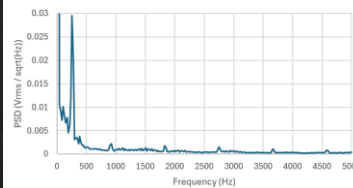
Frequency: 214.84
Power Level: 0.121
A3

Frequency: 214.84,
429.69, 666.02
Power Level: 0.064,
0.006, 0.007

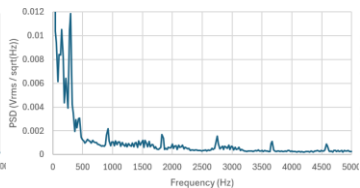
Frequency: 226.33
Power Level: 0.107
B3

Frequency: 236.33,
494.14, 730.47
Power Level: 0.079,
0.029, 0.006

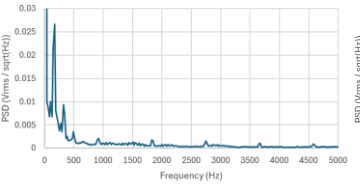
Microphone C3 Power Spectrum



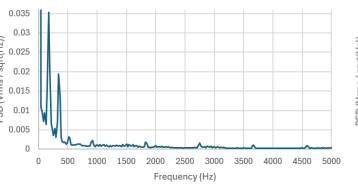
Microphone D3 Power Spectrum



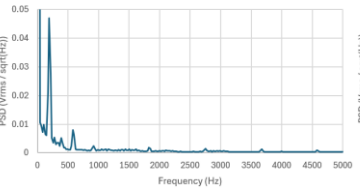
Microphone E3 Power Spectrum



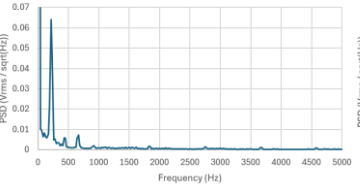
Microphone F3 Power Spectrum



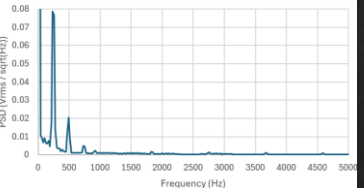
Microphone G3 Power Spectrum



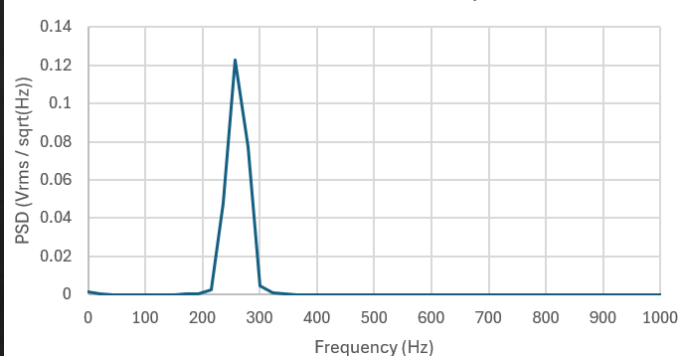
Microphone A3 Power Spectrum



Microphone B3 Power Spectrum



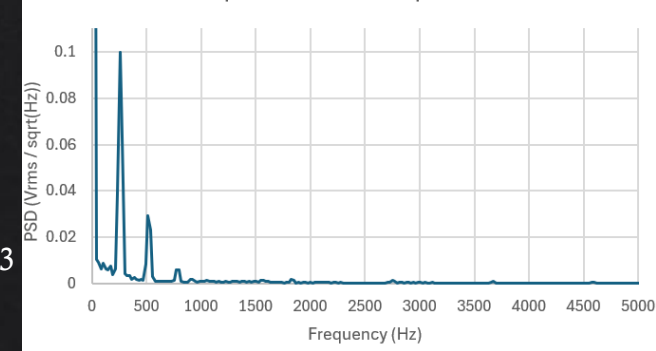
Audio Jack C4 Power Spectrum



Audio Jack
Frequency: 257.81
Power Level: 0.122536

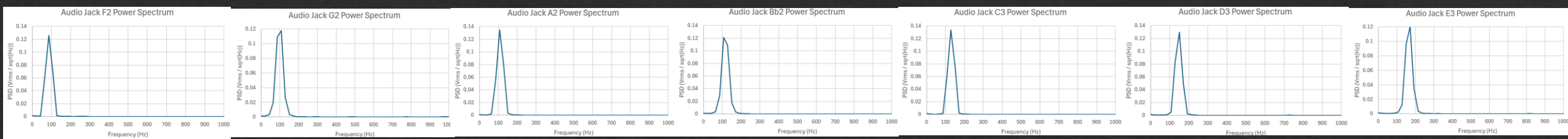
Microphone
Frequency: 257.81, 515.63, 794.92
Power Level: 0.09957, 0.02926, 0.00593

Microphone C4 Power Spectrum



Frequency (Hz)
Power Level (Vrms/ $\sqrt{\text{Hz}}$)

Week 3 Results – F Major Scale



Frequency: 85.94
Power Level: 0.126
F2
Frequency: 85.94,
181.88, 257.81
Power Level: 0.008,
0.028, 0.007

Frequency: 107.42
Power Level: 0.117
G2
Frequency: 107.42,
193.36, 300.78
Power Level: 0.09,
0.020, 0.009

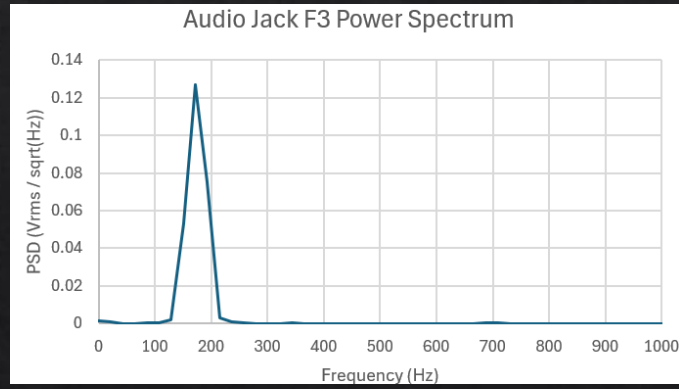
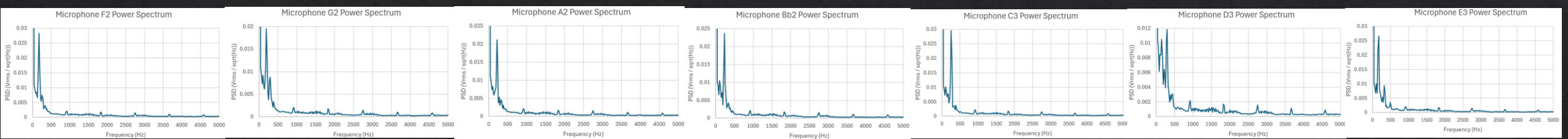
Frequency: 107.42
Power Level: 0.135
A2
Frequency: 107.42,
193.36, 300.78
Power Level: 0.008,
0.021, 0.004

Frequency: 107.42
Power Level: 0.121
Bb2
Frequency: 107.42,
236.33, 343.75
Power Level: 0.010,
0.024, 0.004

Frequency: 128.91
Power Level: 0.134
C3
Frequency: 107.42,
257.81, 408.20
Power Level: 0.010,
0.030, 0.003

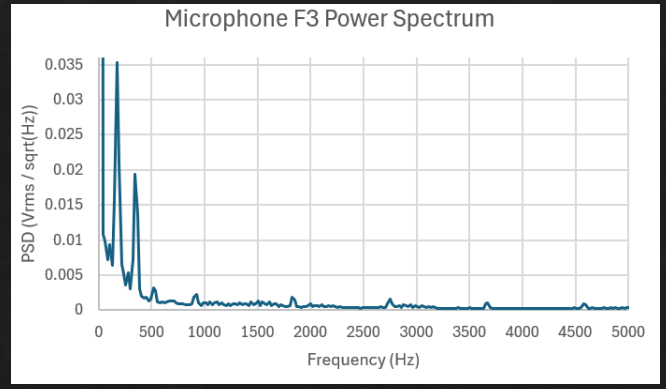
Frequency: 150.39
Power Level: 0.130
D3
Frequency: 150.29,
300.78, 451.17
Power Level: 0.011,
0.012, 0.003

Frequency: 171.88
Power Level: 0.120
E3
Frequency: 171.88,
322.27, 494.43
Power Level: 0.027,
0.009, 0.004



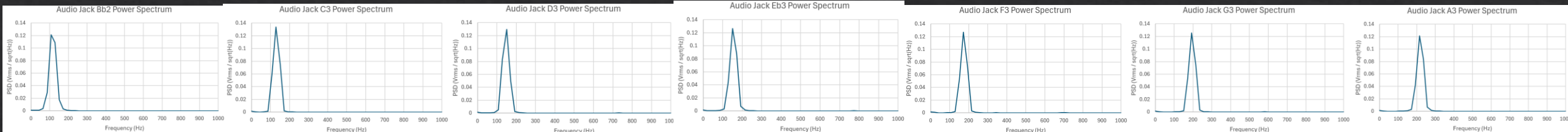
Audio Jack
Frequency: 171.88
Power Level: 0.127086

Microphone
Frequency: 171.88, 343.75, 515.63
Power Level: 0.03526, 0.01938, 0.00324



Frequency (Hz)
Power Level (Vrms/ $\sqrt{\text{Hz}}$)

Week 3 Results – Bb Major Scale



Frequency: 107.42
Power Level: 0.121
Bb2
Frequency: 107.42,
236.33, 343.75
Power Level: 0.010,
0.024, 0.004

Frequency: 128.91
Power Level: 0.134
C3
Frequency: 107.42,
257.81, 408.20
Power Level: 0.010,
0.030, 0.003

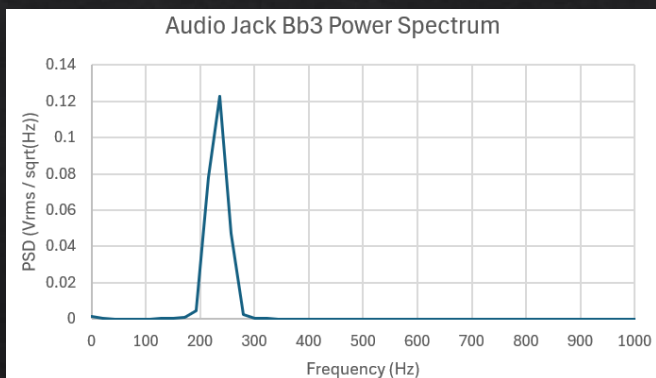
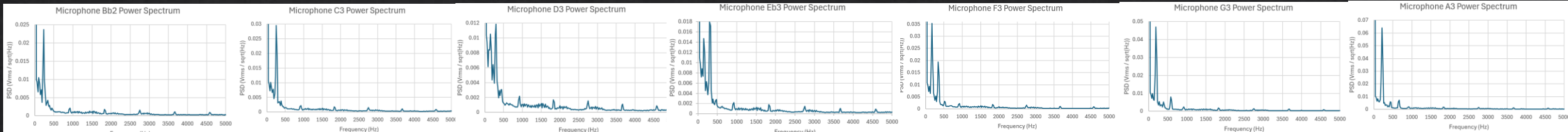
Frequency: 150.39
Power Level: 0.130
D3
Frequency: 150.29,
300.78, 451.17
Power Level: 0.011,
0.012, 0.003

Frequency: 150.39
Power Level: 0.127
Eb3
Frequency: 150.39,
300.78, 472.66
Power Level: 0.015,
0.018, 0.003

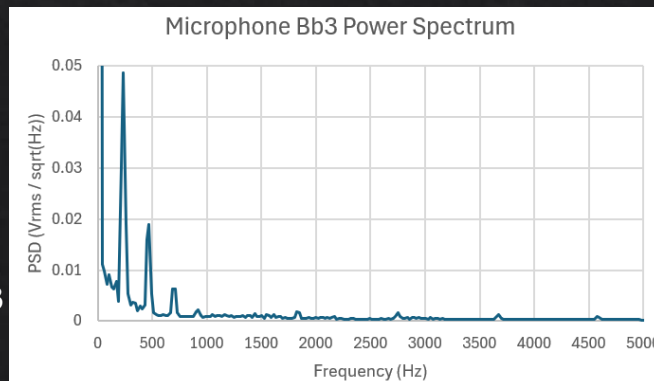
Frequency: 171.88
Power Level: 0.127
F3
Frequency: 171.88,
343.75, 515.63
Power Level: 0.035,
0.019, 0.003

Frequency: 193.36
Power Level: 0.125
G3
Frequency: 193.36,
386.72, 580.08
Power Level: 0.047,
0.005, 0.007

Frequency: 214.84
Power Level: 0.121
A3
Frequency: 214.84,
429.69, 666.02
Power Level: 0.064,
0.006, 0.007



Audio Jack
Frequency: 236.33
Power Level: 0.122547
Bb3
Microphone
Frequency: 236.33, 472.66, 708.98
Power Level: 0.04869, 0.01897, 0.0000633



Errors/Uncertainties

◇ There are a few important errors that could have skewed the results of this experiment.

1. Digital System Acquisition Error

- In all data gathered from the microphone, there was a significant spike in the power spectrum at frequency = 0 Hz.
- The connection of the audio jack to my computer/audio output device may have been altered throughout the course of the experiment.

2. Audio Output Device Error

- The laptop speaker used as the audio output device may not have been able to perfectly replicate the sounds created by the tone generator.
- The tone generator was unable to properly differentiate the frequency between a few of the most similar notes. (D & Eb)

3. Human Error

- Measurement error when recording the specific frequency peaks and power levels.



Week 3 – Conclusion

◆ There are two main takeaways from this experiment.

1. The microphone was able to capture harmonic overtones while the audio jack could not. Overtones simply refer to the higher frequency peaks that occur when a music note is played in real life. When a musical note is played as a digital signal, it is isolated to produce a near-perfect sine wave. The audio jack was able to accurately record a note's frequency but completely overlooked the overtones it created. The microphone was able to pick up these slight reverberations caused by a sound being played in a physical room.

2. White noise is responsible for the significant difference in relative power level between the data recorded by the audio jack and microphone. The audio jack was virtually untouched by white noise, while the microphone was heavily impacted because it pick up information from all the sounds created in the small room. As a result, the peak frequencies recorded by the audio jack will have a much higher power level compared to the microphone.





The End

